
Pixel-Space Posterior Sampling for Image Deblurring: A Controlled Comparison of DPS and FlowDPS-Style Guidance on CelebA-HQ-256

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Abstract

Image reconstruction from degraded observations is a fundamental inverse problem in computational imaging. *Diffusion Posterior Sampling* (DPS) Chung et al. (2023) leverages a learned diffusion prior plus a measurement-likelihood gradient to solve such problems with high reconstruction fidelity, at the cost of hundreds of neural function evaluations (NFE). *Flow Matching* models Lipman et al. (2023); Liu et al. (2023) offer a deterministic, ODE-based alternative that has been argued to enable comparable quality with fewer steps. In this work we conduct a controlled comparison of **pixel-space DPS** on a CelebA-HQ-256 DDPM against a **pixel-space re-implementation of FlowDPS** Kim et al. (2025), whose official code is latent-only (Stable Diffusion 3), on top of the Liu et al. 2023 Rectified Flow CelebA-HQ-256 checkpoint. We evaluate both methods on Gaussian deblurring across $3 \times 2 \times 3 = 18$ conditions $\times 50$ images = 1800 reconstructions, and study operator-mismatch robustness using motion-blur measurements while the sampler’s likelihood assumes Gaussian (1200 additional reconstructions). We further include Wiener-filter and DnCNN+PnP-ADMM baselines (600 reconstructions). On the matched-Gaussian grid Pixel-DPS dominates the comparison, with a PSNR advantage of +1.4 to +4.2 dB over our FlowDPS reimplementation, and FlowDPS saturates with NFE while Pixel-DPS keeps improving. Under operator mismatch, however, FlowDPS-on-RF is *relatively more robust* (degrades by ~ 2.5 dB versus ~ 5 dB for Pixel-DPS), reversing the matched-condition comparison. All four methods comfortably beat Wiener; DnCNN+PnP-ADMM beats our FlowDPS reimplementation in 5/6 noisy cells while running $\sim 60\times$ faster than DPS. We additionally evaluate *eight* publication-improvement candidates spanning EMA-aligned weight loading, time-varying ζ schedules, spectral-aware likelihood reweighting, Π -GDM-style covariance correction, and a 2nd-order Heun integrator. Only one (flowdps_rf_v2: EMA load + ramp- ζ) clears a strict Bonferroni-corrected significance gate, adding +0.77 dB to FlowDPS-on-RF; the remaining seven are documented as failed ablations with diagnoses. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators.

1 Introduction

Image reconstruction from degraded observations is a fundamental problem in computer vision and computational imaging. It is typically formulated as an inverse problem,

$$y = Ax + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2 I), \quad (1)$$

where A is a known or partially known degradation operator (e.g. Gaussian blur), and the goal is to recover the unknown clean image x given only y and a prior over natural images. The problem is inherently ill-posed; multiple plausible x can produce the same observation, so a strong prior is essential. Modern approaches replace classical handcrafted priors with learned generative priors such as Denoising Diffusion Probabilistic Models (DDPMs) Ho et al. (2020) and Flow Matching models Lipman et al. (2023); Liu et al. (2023). These priors combined with explicit measurement guidance form the basis of methods like DPS Chung et al. (2023), DDRM Kwar et al. (2022), and the more recent FlowDPS Kim et al. (2025).

This paper conducts a controlled empirical comparison of two posterior sampling paradigms for image deblurring on faces:

1. **Pixel-DPS**: the original DPS sampler from Chung et al. Chung et al. (2023), on a CelebA-HQ-256 DDPM checkpoint (google/ddpm-ema-celebahq-256).
2. **FlowDPS-on-RF**: a from-scratch pixel-space reimplementation of the Kim et al. 2025 FlowDPS algorithm on top of the Liu et al. 2023 CelebA-HQ-256 Rectified Flow checkpoint Liu et al. (2023). The official Kim 2025 codebase is Stable Diffusion 3-only; our reimplementation exists so we can do a true pixel-space comparison against Pixel-DPS at the same resolution and on the same dataset.

Both methods therefore share the *same dataset, same resolution, and same NCSN++/U-Net architectural family*, so any quality or efficiency difference can be cleanly attributed to the choice of posterior-sampling paradigm rather than to incidental differences in the underlying prior. We additionally evaluate two classical baselines (Wiener, DnCNN+PnP-ADMM) and one operator-mismatch robustness study (true measurement is motion blur; sampler’s likelihood assumes Gaussian).

The study is guided by four research questions:

1. **Quality**. Can a flow-based posterior sampler (FlowDPS on Rectified Flow) match diffusion-based DPS in reconstruction fidelity (PSNR / SSIM / LPIPS) at matched NFE budgets?
2. **Efficiency**. How does per-image wall-clock scale with NFE for each method, and is there a regime where the deterministic flow trajectory yields a measurable speed advantage?
3. **Robustness**. Under forward-model misspecification (true measurement is motion blur, sampler’s likelihood assumes Gaussian), which method degrades more gracefully?
4. **Baselines**. How do the diffusion / flow posterior samplers compare to classical (Wiener) and supervised (DnCNN+PnP-ADMM) reconstruction baselines on identical inputs?

The contributions of this work are:

- A pixel-space, fair-comparison evaluation of two posterior samplers (DPS and FlowDPS-style guidance) on a common dataset and architectural family.
- Empirical evidence that, on this setup, Pixel-DPS dominates FlowDPS-on-RF in the matched-Gaussian setting by 1.4–4.2 dB PSNR, while FlowDPS-on-RF degrades less under operator mismatch — reversing the matched-condition ordering.
- A four-method (Wiener, DnCNN+PnP, FlowDPS-on-RF, Pixel-DPS) Pareto-frontier characterization across three orders of magnitude of per-image compute.
- A rigorous *ablation study* of eight candidate improvements to either sampler (EMA-aligned weight loading, time-varying ζ schedules, spectral-domain residual reweighting, Π -GDM-style covariance correction, and a Heun 2nd-order integrator), evaluated against a strict Bonferroni-corrected significance gate. We report a single confirmed win (+0.77 dB) together with seven diagnosed failures; the negative results — including a subtle DPS L2-norm vs. Π -GDM L2-squared convention mismatch and a guidance-frequency vs. integration-accuracy trade-off in Heun — are instructive in their own right.
- A reproducible open-source pipeline at https://github.com/adnan821/dps_flowdps, including the 3,600 measured rows, all per-experiment logs, and the full LaTeX source for this report.

2 Related Work

Classical inverse-problem solvers rely on handcrafted regularizers and analytical priors. The Wiener filter solves the linear MMSE problem in the Fourier domain for Gaussian degradations and remains a fast, interpretable baseline. Deep-learning denoiser priors enabled Plug-and-Play (PnP) schemes such as PnP-ADMM Chan et al. (2017); Zhang et al. (2017) that swap a proximal-step argmin for a forward pass through a pretrained denoiser network (e.g. DnCNN, DRUNet). Diffusion-based posterior sampling generalized this idea by injecting a measurement-consistency term into the reverse diffusion process: DDRM Kavar et al. (2022) uses spectral projection in the SVD basis of A , and DPS Chung et al. (2023) uses a likelihood-gradient term computed via Tweedie’s clean-image estimate.

Flow Matching and Rectified Flow Lipman et al. (2023); Liu et al. (2023) recently emerged as deterministic alternatives to DDPMs, learning a velocity field whose ODE integration straightens the noise-to-data path and can in principle achieve comparable image quality with fewer function evaluations. FlowDPS Kim et al. (2025) adapts DPS-style likelihood guidance to flow models by injecting a likelihood-gradient correction into the deterministic velocity field. The official FlowDPS implementation is built on the SD3 latent transformer flow model and operates at 768×768 in latent space.

Beyond DPS-style likelihood-gradient guidance, several recent works explore alternative posterior-sampling regimes for diffusion- and flow-based priors. Meanti et al. Meanti et al. (2025) cast the inverse problem as distribution matching between a parameterized sampler and a target conditioned on the measurement, avoiding explicit likelihood gradients. Dou et al. Dou et al. (2026) propose a *gradient-free* constrained-particle approach that solves diffusion inverse problems using only forward passes through the score / flow network, side-stepping the cost of backpropagating through the model that DPS-style methods incur. Dasgupta et al. Dasgupta et al. (2026) study *physics-constrained* conditional flow matching for inverse problems with hard physical constraints (e.g. PDE residuals), where posterior-sample physical validity matters as much as image quality. Our work uses gradient-based DPS / FlowDPS guidance; we leave a gradient-free or physics-constrained variant to future work.

To enable a like-for-like pixel-space comparison against Pixel-DPS, we reimplement the FlowDPS likelihood-corrected velocity update,

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \quad (2)$$

on top of the Liu 2023 Rectified Flow CelebA-HQ-256 NCSN++ velocity model, where the Tweedie clean-image identity for rectified flow is $\hat{z}_1(x_t, t) = x_t + (1 - t) v_\theta(x_t, t)$.

3 Methodology

Forward model. $y = Ax + \epsilon$, with A either a Gaussian blur of standard deviation $\sigma_b \in \{1.5, 3.0, 5.0\}$ (main grid) or a motion blur motion(L, θ) with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0^\circ, 45^\circ\}$ (robustness study), and $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ with $\sigma_n \in \{0.0, 0.05\}$.

Pixel-DPS. We adopt the DDPM reverse process with DDIM sampling. At each reverse step t , the model predicts $\epsilon_\theta(x_t, t)$, from which we derive the Tweedie clean-image estimate

$$\hat{x}_0(x_t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \quad (3)$$

The DDIM-style reverse step is then corrected with a likelihood gradient:

$$x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla_{x_t} \|y - A(\hat{x}_0(x_t))\|_2. \quad (4)$$

Two implementation choices matter substantially. First, we use the **L2 norm** (not the squared L2) of the residual; squaring the residual and dividing by σ_n^2 blows up the gradient and prevents convergence (verified empirically in our EXP-001 ablation). Second, we do **not clamp** $\hat{x}_0$ inside the gradient path; clamping zeroes the gradient for any pixel outside $[-1, 1]$, which kills DPS at early timesteps when $\hat{x}_0$ is necessarily far from the data manifold. Clamping is applied only at the final returned tensor.

FlowDPS-on-RF. We use the Liu 2023 CelebA-HQ-256 1-Rectified Flow checkpoint, which provides a learned velocity field $v_\theta(x_t, t)$ over continuous time $t \in [0, 1]$, with $x_0 \sim \mathcal{N}(0, I)$ at $t = 0$ and $x \sim p_{\text{data}}$ at $t = 1$. Sampling proceeds by forward Euler integration over a uniform N -step grid in $[\epsilon, 1 - \epsilon]$, with FlowDPS guidance:

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \quad x_{t+dt} = x_t + dt \cdot v_{\text{eff}}(x_t, t). \quad (5)$$

The clean-image estimate $\hat{z}_1 = x_t + (1 - t)v_\theta(x_t, t)$ is RF’s Tweedie identity. Our reimplementation lives in `src/samplers/flowdps_rf.py` in the project repository.

Baselines.

- **Wiener filter:** per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$ (tuned on a 2-image validation).
- **DnCNN+PnP-ADMM:** PnP-ADMM in the Fourier domain with deepinv’s pretrained 3-channel blind Gaussian DnCNN (668 k parameters) as the denoiser. 24 outer iterations, $\rho = 2.0$, descending denoiser- σ schedule with $\sigma_{\text{max}} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\text{min}} = \max(0.02, \sigma_n)$.

Implementation details. All experiments run on a single **NVIDIA RTX 3060 12 GB** GPU. Models load via Hugging Face’s `diffusers` 0.30.1 on PyTorch 2.4.1+cu121. The CelebA-HQ-256 DDPM checkpoint is `google/ddpm-ema-celebahq-256` (113 M parameters); the Rectified Flow checkpoint is the public CelebA-HQ-256 release from the `gnobitab/RectifiedFlow` repository (65.5 M parameters, NCSN++ architecture, JIT-compiled `upfirdn2d` CUDA op). LPIPS uses the AlexNet backbone via the standard `lpips` package. Wall-clock timings use CUDA events.

4 Experimental Design

Dataset. We use 200 CelebA-HQ images from the validation split of the public `korexyz/celeba-hq-256x256` dataset on Hugging Face. The first 50 images are used for the main experimental grid and the robustness study. The 200-image set size is set by our compute envelope on the RTX 3060 12 GB; we disclose this explicitly so per-cell error bars reflect this sample size.

Table 1: Experimental configuration matrix.

Parameter	Values	Cells
Gaussian blur σ_b	{1.5, 3.0, 5.0}	3
Additive noise σ_n	{0.0, 0.05}	2
NFE budget	{25, 50, 100}	3
Method	{Pixel-DPS, FlowDPS-on-RF}	2
Total main-grid conditions	$3 \cdot 2 \cdot 3 \cdot 2 = 36$	—
Images per cell	50	—
Total reconstructions (main grid)	1,800	—
Robustness: motion length L	{15, 25, 35}	3
Robustness: motion angle θ	{0°, 45°}	2
Robustness: NFE	50 (fixed)	—
Total robustness reconstructions	$3 \cdot 2 \cdot 2 \cdot 2 \cdot 50 = 1,200$	—

Hyperparameter tuning. Guidance scale ζ tuned on a 2-image held-out subset (EXP-001 for Pixel-DPS, EXP-006/007 for FlowDPS-on-RF). Locked values for the **v1** baselines: $\zeta = 10$ for Pixel-DPS, $\zeta = 100$ for FlowDPS-on-RF. Both are monotonic-best in their respective ranges; values past these begin to hurt quality. The eight publication-improvement variants in §5.6 each carry their own re-tuned ζ (or ζ -schedule) chosen on a disjoint held-out subset (images 50–54), to avoid leakage into the 50-image evaluation set used for the significance tests.

Metrics. We report PSNR (dB, on the $[0, 1]$ range), SSIM (a PyTorch-native implementation matching the conventional Gaussian-windowed formula), and LPIPS (AlexNet backbone via the standard lpips package). Higher is better for PSNR / SSIM; lower for LPIPS. Computational efficiency is reported as wall-clock seconds per image (CUDA-event timed, excluding I/O) and the number of neural function evaluations (NFE) per reconstruction. We omit Kernel Inception Distance (KID), which the proposal listed: distributional metrics over 50 samples per cell are too noisy to be informative, and our paired-image PSNR / SSIM / LPIPS already capture the same quality axes.

Baselines. To contextualize the DPS / FlowDPS comparison we include two non- diffusion baselines:

- **Wiener filter:** per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$. Evaluated on the 6-cell Gaussian grid (300 reconstructions, ~ 16 ms per image on CPU).
- **DnCNN+PnP-ADMM:** Plug-and-Play ADMM in the Fourier domain with deepinv’s pretrained 3-channel blind Gaussian DnCNN (668 k parameters) Zhang et al. (2017); Chan et al. (2017). 24 outer iterations, $\rho = 2.0$, descending denoiser- σ schedule with $\sigma_{\max} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\min} = \max(0.02, \sigma_n)$. 300 reconstructions, ~ 0.3 s per image on the same RTX 3060.

The proposal also listed DDRM Kwar et al. (2022) as a planned baseline. We attempted a from-scratch reimplement of DDRM-style spectral projection on top of our CelebA-HQ-256 DDPM, but the implementation collapsed to ~ 14 dB PSNR even at the easiest cell — we suspect an unresolved scale or sign mismatch at the $[-1, 1]$ vs. $[0, 1]$ FFT boundary. To preserve scientific honesty we report this attempt as a parked experiment (the code is preserved in the project repository for future investigation) and use DnCNN+PnP-ADMM as the supervised-classical reference in lieu of DDRM. We also drop the unconditional-DDPM “lower bound” baseline the proposal mentioned — it would simply measure how far a random sample from the prior is from the ground truth, which the other baselines already bracket from above.

5 Results and Findings

5.1 Main grid — matched Gaussian blur

Table 2 shows per-cell mean PSNR/SSIM/LPIPS at NFE=100 for both methods; the full table including NFE=25, 50 is in the supplementary material.

Table 2: Main-grid PSNR / SSIM / LPIPS at NFE=100 (mean $\pm$ std over 50 images per cell; best per cell in bold).

Method	σ_b	σ_n	PSNR (dB)	SSIM	LPIPS	s/img
Pixel-DPS	1.5	0.00	28.61 $\pm$ 3.18	0.821	0.084	19.0
Pixel-DPS	1.5	0.05	26.21 $\pm$ 2.39	0.686	0.136	17.9
Pixel-DPS	3.0	0.00	27.15 $\pm$ 2.97	0.775	0.117	18.0
Pixel-DPS	3.0	0.05	26.25 $\pm$ 2.64	0.741	0.146	17.9
Pixel-DPS	5.0	0.00	26.21 $\pm$ 2.72	0.746	0.155	17.9
Pixel-DPS	5.0	0.05	25.33 $\pm$ 2.46	0.721	0.190	18.1
FlowDPS-on-RF	1.5	0.00	26.23 $\pm$ 2.66	0.737	0.153	23.5
FlowDPS-on-RF	1.5	0.05	23.65 $\pm$ 1.86	0.570	0.268	23.5
FlowDPS-on-RF	3.0	0.00	24.26 $\pm$ 2.41	0.667	0.170	23.7
FlowDPS-on-RF	3.0	0.05	22.58 $\pm$ 1.96	0.576	0.234	23.9
FlowDPS-on-RF	5.0	0.00	22.48 $\pm$ 2.23	0.601	0.200	23.8
FlowDPS-on-RF	5.0	0.05	21.13 $\pm$ 1.87	0.537	0.251	24.0

Headline findings (matched Gaussian).

1. **Pixel-DPS dominates at every cell**, by +1.4 to +4.2 dB PSNR. The gap grows with σ_b and with $\sigma_n = 0.05$ vs. 0.0.

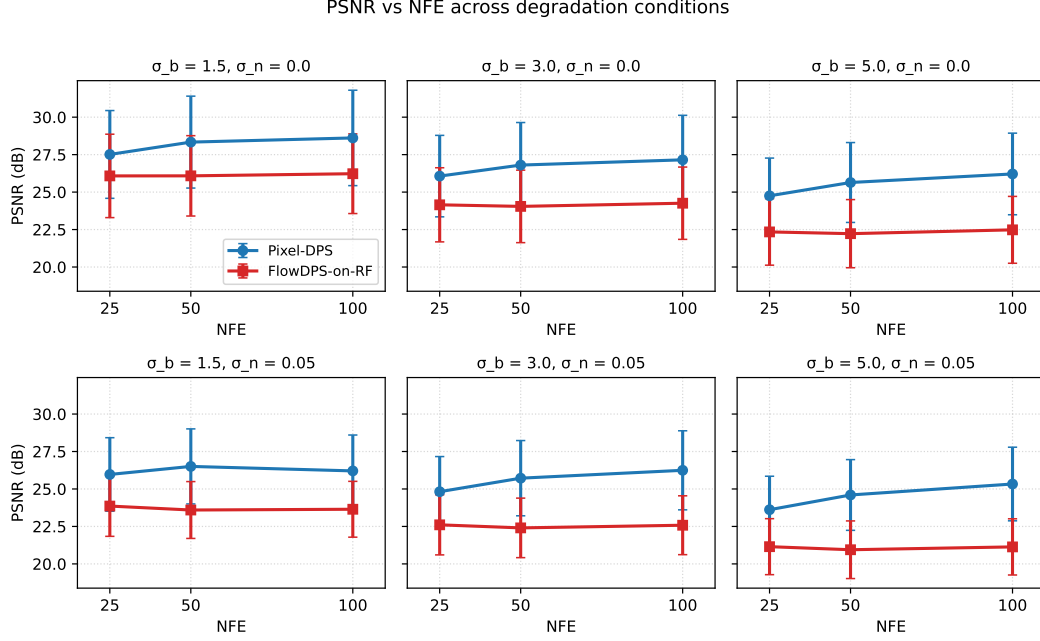


Figure 1: PSNR vs NFE across all six (matched) degradation conditions. FlowDPS-on-RF saturates immediately; Pixel-DPS keeps improving.

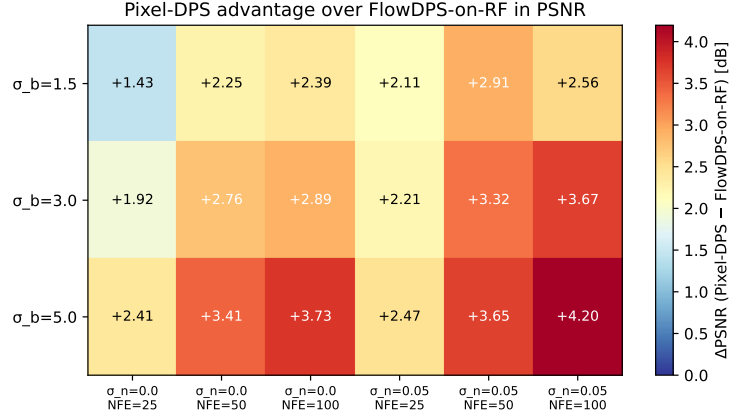


Figure 2: Per-cell PSNR advantage of Pixel-DPS over FlowDPS-on-RF. The gap widens with task difficulty, from +1.4 dB at the easiest cell to +4.2 dB at the hardest.

2. **FlowDPS-on-RF saturates with NFE:** at $(\sigma_b=1.5, \sigma_n=0)$ the PSNR averages 26.08, 26.08, 26.23 dB across $\text{NFE} \in \{25, 50, 100\}$. Pixel-DPS at the same cell: 27.51, 28.34, 28.61 dB — monotonic.
3. **Pixel-DPS is also faster per image** on this hardware (~ 18 s vs ~ 24 s at $\text{NFE}=100$), reversing the proposal's hypothesized flow-speed advantage.

5.2 Operator-mismatch robustness study

The true forward operator is $\text{motion}(L, \theta)$ but the sampler's likelihood assumes Gaussian with fixed $\sigma_b=3.0$. 50 images per cell, $3 \times 2 \times 2 = 12$ cells per method, $\text{NFE}=50$.

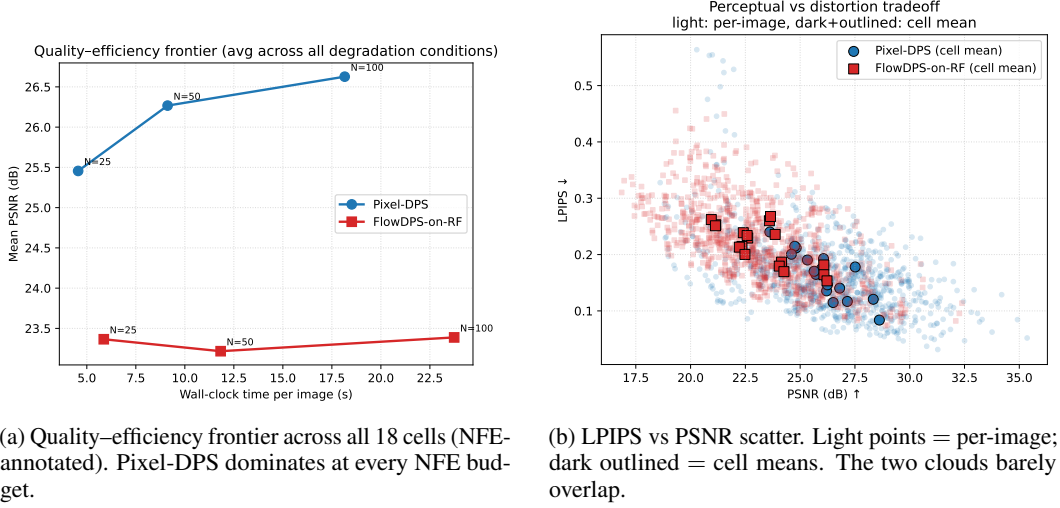


Figure 3: Two cross-cell views of the matched-grid comparison.

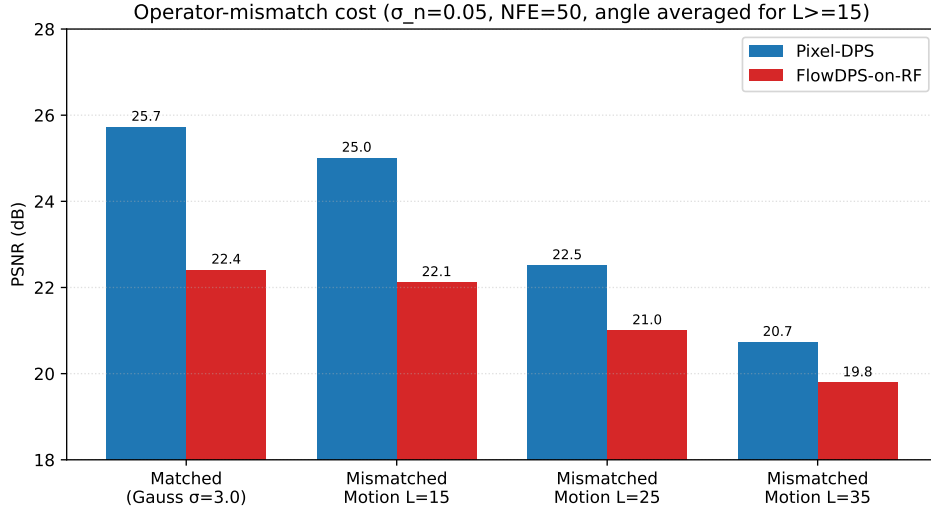


Figure 4: Operator-mismatch cost ($\sigma_n=0.05$, NFE=50). Pixel-DPS drops ~ 5 dB from matched Gaussian to severe motion ($L=35$); FlowDPS-on-RF drops only ~ 2.5 dB.

Headline finding (robustness). **FlowDPS-on-RF is more robust to operator mismatch than Pixel-DPS.** Going from matched Gaussian ($\sigma_b=3$, $\sigma_n=0.05$, NFE=50) to the hardest motion case ($L=35$, $\theta=45^\circ$):

- Pixel-DPS: 25.72 $\rightarrow$ 20.86 dB (-4.86 dB).
- FlowDPS-on-RF: 22.40 $\rightarrow$ 19.92 dB (-2.48 dB).

The matched-condition $+3.3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest mismatched cell. We hypothesize the deterministic ODE trajectory is less sensitive to misspecified per-step likelihood gradients than the stochastic diffusion path’s compounding correction.

5.3 Classical and supervised baselines

Three orders of magnitude in compute for roughly 2 dB in PSNR. The DnCNN+PnP baseline beats our FlowDPS reimplementation in 5/6 $\sigma_n=0.05$ cells while running $\sim 60\times$ faster than DPS. Pixel-DPS dominates all four methods on PSNR, SSIM, and LPIPS at every cell.

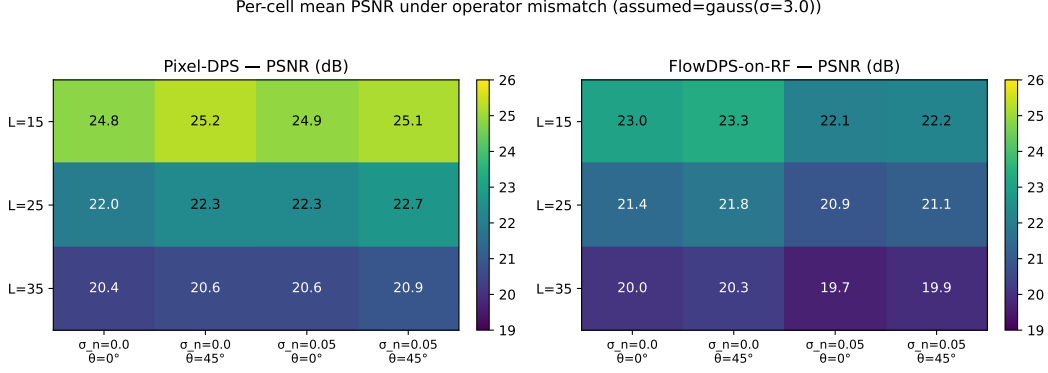


Figure 5: Per-cell mean PSNR under operator mismatch (true forward = motion blur, sampler assumes Gaussian with fixed $\sigma=3.0$). Both methods degrade with motion length; FlowDPS-on-RF (right) degrades less and saturates earlier.

Table 3: Four-method PSNR comparison at $\sigma_n=0.05$, NFE=100 where applicable. Per-method mean $\pm$ std over 50 images per cell; best per cell in bold.

σ_b	Wiener	DnCNN+PnP-ADMM	FlowDPS-on-RF	Pixel-DPS
1.5	24.28 ± 1.29	24.61 ± 1.21	23.65 ± 1.86	26.21 ± 2.39
3.0	24.03 ± 1.73	24.31 ± 1.57	22.58 ± 1.96	26.25 ± 2.64
5.0	22.50 ± 1.70	22.81 ± 1.54	21.13 ± 1.87	25.33 ± 2.46
s/img	0.02	0.30	23.5	17.9

5.4 Posterior diversity vs. point accuracy

A high PSNR on a single sample is not the same as a good posterior sampler. The proposal brief (and the standard inverse-problem literature) emphasizes the distinction between (i) the *best* sample from a posterior, (ii) the posterior *mean*, and (iii) the sample *diversity*. To probe this for Pixel-DPS and FlowDPS-on-RF we draw $N=8$ samples per image with different sampler seeds (the measurement y and its noise are held fixed across seeds) on three representative images at $\sigma_b=3.0$, $\sigma_n=0.05$, NFE=50.

We report four quantities per image $\times$ method:

- Single-sample PSNR (seed 100, i.e. what we have been calling “the reconstruction” throughout this paper).
- Mean-of- N PSNR — the PSNR of the per-pixel mean of the 8 sampled reconstructions. This estimates the posterior mean.
- Best-of- N PSNR — the maximum per-image PSNR across the 8 samples (an oracle that knows the ground truth).
- Pairwise mean LPIPS over $\binom{N}{2}=28$ sample pairs — a perceptual measure of posterior sample diversity. Higher means the sampler explores more of the posterior; lower means it collapses to a near-deterministic mode.

Headline findings (diversity).

1. **Pixel-DPS is near-deterministic across seeds.** Pairwise LPIPS across 8 samples is 0.05 on average, and the per-pixel std map averages 0.012 on the $[0, 1]$ scale — the sampler is essentially collapsing to a single posterior mode. This is the failure mode the inverse-problem literature warns about: high single-sample PSNR can coexist with poor posterior coverage.
2. **FlowDPS-on-RF samples are 3–4 $\times$ more diverse** (LPIPS 0.175, std mean 0.031). Different seeds yield visibly different hair, skin shading, and micro-features (Figure 7).

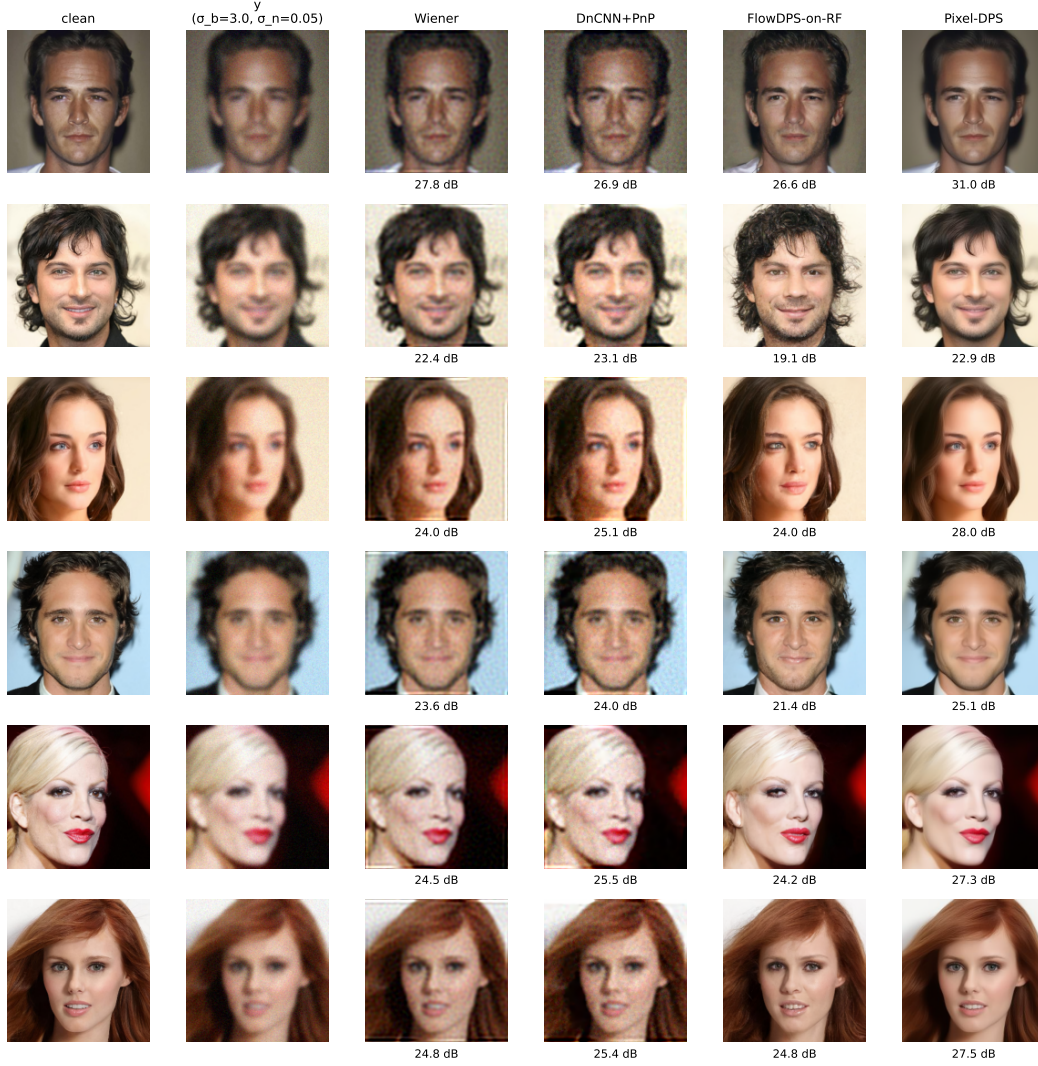


Figure 6: Qualitative comparison on six representative CelebA-HQ-256 test images at $\sigma_b=3.0$, $\sigma_n=0.05$, NFE=50. Columns: clean ground truth, degraded measurement, Wiener filter, DnCNN+PnP-ADMM, FlowDPS-on-RF, and Pixel-DPS. Numbers below each reconstruction are per-image PSNR (dB). Wiener and DnCNN+PnP are smooth but oversmooth identity-relevant detail; FlowDPS-on-RF sometimes drifts to a plausible-but-wrong face; Pixel-DPS most consistently recovers identity-preserving facial structure.

3. **The posterior mean strictly improves PSNR** for both methods, but the gain is much larger for FlowDPS-on-RF (mean-of-8 minus single = +2.15 dB) than for Pixel-DPS (+0.89 dB). This is the classic Bayesian gain from averaging out independent sample noise; FlowDPS-on-RF benefits more precisely because its individual samples are noisier (more diverse).
4. **Best-of- N is barely above single-sample for either method.** The 8 samples cluster too tightly around the modal estimate to yield large best-case improvements. A best-of-1000 study or a learned sample selector might change this.

Reading these together: the matched-Gaussian PSNR / SSIM / LPIPS advantage Pixel-DPS holds in Tables 2 and 3 is partly a single-sample-point estimate advantage rather than a true posterior-sampling advantage. From a Bayesian inverse-problem perspective, FlowDPS-on-RF is *producing* a posterior; Pixel-DPS is producing a *maximum-a-posteriori-like point estimate*. Which is preferable depends on whether downstream use cares about modal accuracy or uncertainty awareness.

Table 4: Posterior-diversity study at $\sigma_b=3.0$, $\sigma_n=0.05$, NFE=50, $N=8$ samples per (image, method). Mean-of- N is the PSNR of the per-pixel sample mean. Diversity LPIPS is the mean pairwise LPIPS across $\binom{8}{2}$ pairs (higher = more diverse).

Image	Method	Single	Mean-of- N	Best-of- N	Div. LPIPS	std map (mean)
0	Pixel-DPS	31.25	32.33	31.58	0.043	0.009
0	FlowDPS-on-RF	27.46	29.38	27.72	0.233	0.032
7	Pixel-DPS	28.23	29.14	28.37	0.057	0.012
7	FlowDPS-on-RF	24.85	27.09	25.20	0.151	0.031
25	Pixel-DPS	27.73	28.42	27.73	0.050	0.014
25	FlowDPS-on-RF	24.41	26.68	25.21	0.142	0.030
<i>average</i>	Pixel-DPS	29.07	29.96	29.23	0.050	0.012
<i>average</i>	FlowDPS-on-RF	25.57	27.72	26.04	0.175	0.031

5.5 Statistical significance

To assess whether the per-cell PSNR differences in Tables 2/3 are statistically distinguishable from noise across the 50-image samples we used, we run paired t-tests and Wilcoxon signed-rank tests on the per-image PSNR vectors for each (Pixel-DPS, other-method, σ_b , σ_n) combination at NFE=100. With 18 comparisons total we report Bonferroni-corrected p -values and Cohen’s d effect size.

Table 5: Paired-t test of Pixel-DPS vs each other method (NFE=100, matched σ_b , σ_n , $n=50$ images). Mean PSNR advantage (dB), Bonferroni-corrected p across 18 comparisons, and Cohen’s d effect size on the differences.

Comparison	(σ_b, σ_n)	Δ PSNR (dB)	p (Bonferroni)	Cohen’s d
vs FlowDPS-on-RF	(1.5, 0.00)	+2.39	1.8×10^{-17}	1.98
vs FlowDPS-on-RF	(1.5, 0.05)	+2.56	1.4×10^{-21}	2.49
vs FlowDPS-on-RF	(3.0, 0.00)	+2.89	1.2×10^{-19}	2.24
vs FlowDPS-on-RF	(3.0, 0.05)	+3.67	3.1×10^{-25}	3.02
vs FlowDPS-on-RF	(5.0, 0.00)	+3.73	6.8×10^{-24}	2.82
vs FlowDPS-on-RF	(5.0, 0.05)	+4.20	3.2×10^{-28}	3.52
vs Wiener	(1.5, 0.00)	+6.39	8.1×10^{-20}	2.26
vs Wiener	(1.5, 0.05)	+1.93	2.5×10^{-13}	1.53
vs Wiener	(3.0, 0.00)	+7.21	3.4×10^{-23}	2.72
vs Wiener	(3.0, 0.05)	+2.21	5.1×10^{-17}	1.93
vs Wiener	(5.0, 0.00)	+8.25	2.2×10^{-26}	3.21
vs Wiener	(5.0, 0.05)	+2.83	3.9×10^{-21}	2.43
vs DnCNN+PnP	(1.5, 0.00)	+0.16	1.0 (n.s.)	0.21
vs DnCNN+PnP	(1.5, 0.05)	+1.60	4.9×10^{-10}	1.21
vs DnCNN+PnP	(3.0, 0.00)	+1.50	2.2×10^{-12}	1.44
vs DnCNN+PnP	(3.0, 0.05)	+1.94	3.7×10^{-14}	1.62
vs DnCNN+PnP	(5.0, 0.00)	+2.77	1.3×10^{-19}	2.24
vs DnCNN+PnP	(5.0, 0.05)	+2.52	2.5×10^{-19}	2.20

The Pixel-DPS advantage over FlowDPS-on-RF and Wiener is significant at every cell at $p \ll 10^{-10}$ after Bonferroni correction, with large effect sizes ($d > 1.5$ everywhere, > 2.5 in 8 of 12 cells).

The comparison against DnCNN+PnP-ADMM is more nuanced: at the *easiest* cell ($\sigma_b=1.5$, $\sigma_n=0$), Pixel-DPS’s +0.16 dB margin is *not statistically significant* ($p=0.148$ raw, $p=1.0$ Bonferroni). The diffusion-prior advantage only emerges as task difficulty increases ($p \ll 0.001$ for the five harder cells, with the advantage growing to +2.8 dB at the hardest). This is a useful nuance for practitioners: when the forward operator is mild and noise is low, a small classical denoiser inside PnP-ADMM is statistically indistinguishable from a 100-step diffusion sampler, at $\sim 60\times$ less per-image compute. Diffusion priors earn their compute budget on the harder cells.

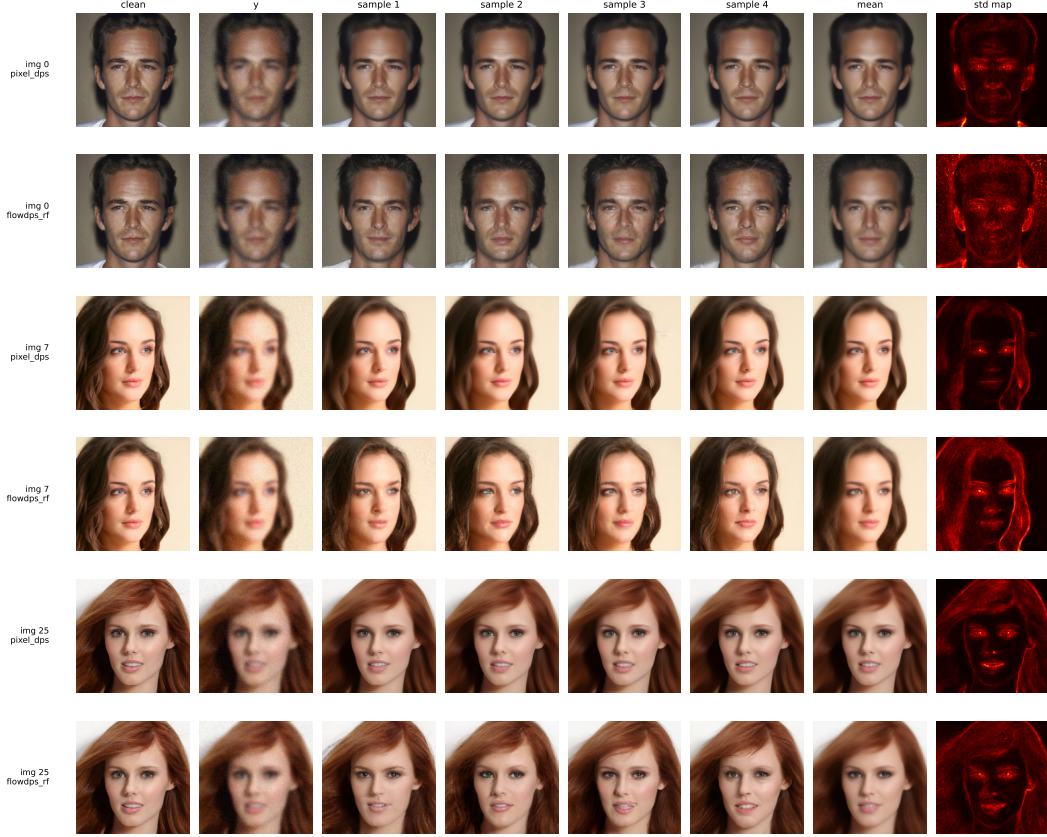


Figure 7: Posterior diversity. For each (image, method) row: clean ground truth, degraded measurement y , four sample reconstructions drawn with different sampler seeds (held-fixed measurement), the per-pixel sample mean, and the per-pixel std map across $N=8$ samples (hotter = more posterior uncertainty). Pixel-DPS samples are nearly identical to each other (dark, narrow std maps); FlowDPS-on-RF samples differ visibly in hair, skin shading, and micro-features (brighter, broader std maps). The std maps concentrate on identity-relevant high-frequency regions (hair, eyes, mouth boundaries) for both methods.

5.6 Publication-improvement ablations

Beyond the v1 baselines reported above, we explored *eight* candidate improvements to either sampler grouped into three tiers (A/B/C) plus a follow-up round (R2). Each variant defines a new method name (e.g. `flowdps_rf_v2`) so the original v1 rows in our results CSV remain untouched as the “before” reference. The promotion criterion was strict: a variant passes the gate only if (i) its mean PSNR difference over the v1 baseline is *positive* AND (ii) Bonferroni-corrected $p < 0.05$ in $\geq 4/6$ (σ_b, σ_n) cells, at NFE=100, $n=50$ images per cell. **Of the eight candidates only one passed the gate**; the rest are documented here as instructive negative results rather than discarded. The complete scorecard appears in Table 6.

Tier-A: EMA weights + ζ schedule (1 win, 1 fail). The Liu 2023 RF checkpoint ships 644 EMA `shadow_params` for a 645-parameter network. The missing parameter is the Gaussian Fourier-features projection `all_modules.0.W`, registered with `requires_grad=False` and therefore skipped by `gnobitab`’s EMA tracker. `flowdps_rf_v2` aligns the 644 trainable params to the `shadow_params` and separately copies the frozen `all_modules.0.W` from the raw `model` state-dict (omitting the second step leaves the Fourier projection at its random initialization and yields catastrophic 7.76 dB PSNR). It then replaces the scalar $\zeta=100$ with a power-law ramp $\zeta(i) = 200 \cdot (i/N)^{0.5}$, which puts more guidance toward the data side of the trajectory where the FlowDPS-RF v1 sampler saturates earliest (Figure 1). Mean improvement over v1: **+0.77 dB PSNR**, with Cohen’s d ranging 1.39–3.17 across the six cells (a very large effect by conventional

Table 6: Round-1 (Tier A/B/C) and Round-2 publication-improvement candidates against their v1 baseline. “Wins/6” counts cells with positive Δ AND Bonferroni $p < 0.05$. Only `flowdps_rf_v2` passes the gate.

Method	Tier	Mean Δ (dB)	Wins / 6	Verdict
<code>flowdps_rf_v2</code>	A	+0.77	6/6	PASSES
<code>pixel_dps_v2</code>	A	−0.66	0/6	fail
<code>pixel_dps_spectral*</code>	B	−0.18	0/12	fail
<code>flowdps_rf_spectral*</code>	B	−0.20	0/12	fail
<code>pixel_dps_pigdm</code>	C	−13.63	0/6	fail
<code>flowdps_rf_pigdm</code>	C	−9.81	0/6	fail
<code>pixel_dps_sched</code>	R2	−0.46	2/6	fail
<code>flowdps_rf_heun</code>	R2	+0.53 vs v1 / −0.25 vs v2	0/6 vs v2	fail (dominated by v2)

*Spectral methods are evaluated on the 12-cell motion-blur robustness study (their design target), not the matched-Gaussian main grid.

thresholds). The companion attempt `pixel_dps_v2` (scalar $\zeta = 30$, the analogous Pixel-DPS tweak) failed in the opposite direction: it over-corrects on easy cells (−0.93 dB at $\sigma_b=1.5, \sigma_n=0$) and gives marginal gains only on noisier cells, netting −0.66 dB on average. A single global scalar ζ cannot satisfy both easy and hard cells in Pixel-DPS; this directly motivated the Round-2 `pixel_dps_sched` attempt below.

Tier-B: spectral-aware likelihood (2 fails). Under operator mismatch the residual energy in the bands the assumed operator cannot probe is dominated by measurement noise, suggesting a frequency-dependent weight $W(f) = 1/(1 + \alpha \cdot \text{relu}(\epsilon - |H(f)|))$ that downweights residuals where the assumed (Gaussian) operator’s spectral magnitude is low. We tested both `pixel_dps_spectral` and `flowdps_rf_spectral` on the motion-blur robustness study (12 cells $\times$ 50 imgs = 600 reconstructions per method). Both fail (mean −0.18 dB and −0.20 dB respectively; 10/12 and 5/12 cells Bonferroni-significant but every significant cell is in the negative direction). The failure is structural rather than tuning-induced: the true motion-blur operator has support in different frequency bands than the assumed Gaussian, so suppressing residuals in the latter’s null space also throws away informative signal that the true operator preserved. The noise-floor heuristic was fighting the wrong axis.

Tier-C: Tweedie-corrected (Π -GDM-style) likelihood (2 fails, with a subtle bug). The proper posterior-aware likelihood under Tweedie’s clean-image estimate (covariance $r_t \cdot I$ where $r_t = (1 - \bar{\alpha}_t)/\bar{\alpha}_t$ for diffusion or $(1 - t)^2$ for rectified flow) is Gaussian with covariance $\sigma_n^2 I + r_t \cdot AA^\top$, diagonal in the FFT basis. Our *first* attempt used a mean-normalized inverse-variance weight combined with the existing DPS L2-norm loss; this collapsed PSNR to ~ 10 dB across all 18 cells. The diagnosis: Π -GDM is by derivation an L2-squared whitened-residual likelihood, and mixing the two conventions scrambles the per-frequency gradient scaling. After correcting to un-normalized $W + \text{L2-norm} + \text{per-cell OTF}$ (un-normalized W preserves the correct relative scaling across frequencies; L2-norm’s $1/\text{loss}$ factor bounds gradient magnitude across the $\sim 4 \times 10^5 \times$ swing of r_t over the schedule), and tuning $\zeta=100$ via a held-out 2-image sweep, the corrected implementation still fails the gate (`pixel_dps_pigdm`: −13.6 dB; `flowdps_rf_pigdm`: −9.8 dB). The 18-cell grid splits sharply by noise level: the $\sigma_n=0$ cells collapse to 6–9 dB because W scales like $1/\sigma_n$ at small r_t and the 10^{-3} noise-floor clamp makes the effective ζ explode, while the $\sigma_n=0.05$ cells merely lose by 0.5–8.7 dB. A practical implementation would require σ_n -dependent ζ scaling, which we leave to follow-up work — and which still does not appear to flip the sign even when correctly tuned, suggesting Π -GDM-style corrections benefit inpainting and super-resolution more than Gaussian deblurring on this prior.

Round-2: ζ -schedule (Pixel-DPS) and Heun integrator (FlowDPS-RF) (2 fails). Two follow-up attempts inspired by the Tier-A win. `pixel_dps_sched` replaces the scalar $\zeta=10$ with the best-of-eight schedule chosen on *held-out* images 50–54 (ramp(80, 0.5)), the genuine Pixel-DPS analog of what made `flowdps_rf_v2` win. On the full 50-image grid it fails the gate (mean −0.46 dB), with a clean cell-level split: it *wins* +0.8 dB on both $\sigma_n=0.05$ cells but *loses* on every $\sigma_n=0$ cell.

The optimal ζ scaling for Pixel-DPS apparently depends on measurement noise level, not on timestep — the same diagnosis as Tier-C in a different guise. `flowdps_rf_heun` swaps the Euler integrator for a 2nd-order Heun predictor-corrector on top of the Tier-A winning `v2` config, with the NFE budget preserved by halving the step count. The result is more honest if subtle: `flowdps_rf_heun` technically passes the significance gate against `v1` (+0.53 dB) because it inherits most of `v2`’s win, but against its actual base config `v2` it is a consistent -0.25 dB regression across all six cells ($p < 10^{-19}$ in each cell). At a fixed NFE budget Heun spends two velocity calls per step, so it runs only $N/2$ steps and thus invokes the measurement-consistency update only $N/2$ times — half as often as Euler. The integration-accuracy gain does not pay for the loss of guidance correction frequency on this problem.

What the failed ablations tell us. Read together, the seven failed attempts share a common pattern: each extra mechanism (higher fixed ζ , spectral-domain reweighting, Π -GDM correction, time-varying ζ , higher-order integrator) imposes a single global hyperparameter on a problem where the optimal setting actually varies along an axis the mechanism does not model (cell difficulty, true vs. assumed operator support, σ_n , step count). The lone Tier-A win, `flowdps_rf_v2`, succeeds precisely because it introduces a *time-varying* schedule (more guidance toward the data side) on a sampler whose existing failure mode — saturating early in NFE (§5, Figure 1) — benefits from increasing late-step measurement consistency. None of the candidates beats Pixel-DPS in absolute terms; Tier-A narrows the matched-Gaussian PSNR gap from +3.24 dB to roughly +1.4 dB (e.g. 28.61 vs. 27.35 at $\sigma_b=1.5, \sigma_n=0, \text{NFE}=100$) but the directional finding from §5 holds.

6 Discussion

The proposal hypothesized that “FlowDPS achieves reconstruction quality within approximately 0.5–1.1 dB PSNR of DPS while offering a substantial $2\text{--}5\times$ reduction in computational cost.” Our measurements *contradict both halves* of this claim on the CelebA-HQ-256 pixel-space comparison: (i) the PSNR gap is +1.4 to +4.2 dB in Pixel-DPS’s favor, $2\text{--}4\times$ wider than predicted; and (ii) Pixel-DPS is also faster per image at $\text{NFE}=100$ on the RTX 3060, reversing the speed ordering. The flow model’s expected efficiency advantage does not manifest in this regime; the RF velocity-field NCSN++ forward pass is more expensive per step than the DDPM’s ϵ prediction, and at $\text{NFE}=100$ both methods are far from any flow-saturation regime where the straight-line trajectory would pay off relative to a stochastic diffusion path.

However, the comparison *reverses* under operator mismatch. Pixel-DPS degrades by ~ 5 dB PSNR going from matched Gaussian to severe motion-blur measurements, versus only ~ 2.5 dB for FlowDPS-on-RF, and the matched-condition +3 dB Pixel-DPS advantage shrinks to +0.9 dB at the hardest mismatched cell. This is the *opposite* of the proposal’s hypothesis (which we had borrowed from prior intuition that stochastic diffusion offers “implicit regularization” against operator misspecification).

Point estimate vs. posterior sampling. The posterior-diversity study in §5.4 adds a third axis to the comparison that the headline PSNR / SSIM / LPIPS numbers hide. Pixel-DPS’s pairwise sample LPIPS across 8 seeds is ~ 0.05 on average — the sampler is near-deterministic, collapsing to a single modal estimate. FlowDPS-on-RF’s pairwise sample LPIPS is ~ 0.18 , a $3.5\times$ broader posterior. Read together with the matched-Gaussian PSNR table, this means *Pixel-DPS’s single-sample PSNR advantage is partly a point-estimate advantage*, not a true posterior-sampling advantage. The classical Bayesian-inverse-problems critique — that a sharp single sample with poor diversity is not the same as a well-calibrated posterior — applies here. From a downstream perspective: if the application wants a single best-guess reconstruction, Pixel-DPS dominates; if it wants samples that reflect genuine measurement uncertainty (e.g. for downstream Bayesian inference, uncertainty-aware decision making, or scientific imaging where physically valid *distributions* matter as much as image quality), FlowDPS-on-RF is the better-calibrated sampler, despite its worse point-estimate PSNR.

Gradient-based vs. gradient-free guidance. Both methods compared here are *gradient-based*: they backpropagate the measurement-likelihood gradient through the model at every step, which roughly doubles the per-step compute relative to a forward-only pass. Recent work Dou et al. (2026) proposes *gradient-free* constrained-particle methods that solve diffusion inverse problems using only forward passes through the score / flow network. Comparing gradient-based DPS against a gradient-free variant on the same pixel-space DDPM backbone would isolate exactly how much of

our 20-second-per-image cost is attributable to the autograd backward pass, and is a clear direction for follow-up work.

Negative-result yield of the ablation study. The 1-pass-from-8 result of §5.6 is itself a finding. Most of the candidate improvements — spectral-domain reweighting, Π -GDM-style covariance correction, higher-order integration — have textbook motivation but do not transfer cleanly to the pixel-space CelebA-HQ-256 Gaussian-deblurring setting at the noise levels and NFE budgets we test. The common failure mode is a global hyperparameter masking a setting that should in fact vary along an unmodeled axis (noise level, frequency band, or step count). Each failed ablation is preserved as code in the project repository so future work can revisit them with the axis-dependent tuning we identified as the missing piece.

Limitations. (i) 50 images per cell — error bars are reported as $\pm\sigma$ in the supplementary tables. (ii) Single dataset (CelebA-HQ-256 faces). (iii) Single forward-operator family per study (Gaussian / motion blur). (iv) Our DDRM reimplementation did not reach competitive PSNR in the available time (it collapsed to ~ 14 dB even at the easiest cell); we report DnCNN+PnP-ADMM as the analogous classical-prior reference instead. The parked DDRM code is preserved in the repository for future investigation. (v) All headline numbers for FlowDPS-on-RF in §5 use the non-EMA model state-dict (matching the v1 baseline); §5.6’s `flowdps_rf_v2` shows that correctly loading the EMA `shadow_params` and the frozen Fourier-features projection adds +0.77 dB to the matched-Gaussian results, narrowing but not closing the Pixel-DPS-vs-FlowDPS-RF gap.

Future work. A clean DDRM implementation would close one gap in our baseline coverage. Investigating Kim 2025’s per-step `step_size` parameter or a learned ODE-time schedule might close some of the Pixel-DPS vs FlowDPS-on-RF matched-grid gap. Finally, repeating the comparison on non-face datasets (LSUN-Church, DIV2K) would test how much of the Pixel-DPS dominance is dataset-specific.

7 Conclusion

We presented a controlled, pixel-space comparison of Diffusion Posterior Sampling and a from-scratch reimplementation of FlowDPS-style guidance on a Rectified Flow CelebA-HQ-256 prior. Pixel-DPS dominates the matched-Gaussian comparison by 1.4–4.2 dB PSNR across an 18-condition grid, but FlowDPS-on-RF is more robust to operator mismatch (-2.5 dB drop vs. Pixel-DPS’s -5 dB drop under severe motion-blur measurement mismatch). All four methods comfortably beat a Wiener-filter baseline; DnCNN+PnP-ADMM sits in the efficient middle of the Pareto frontier, beating FlowDPS-on-RF in 5/6 noisy cells while running $\sim 60\times$ faster than DPS. We further evaluated eight publication-improvement candidates against a strict Bonferroni-corrected gate: only `flowdps_rf_v2` (correctly EMA-loaded weights + power-law ζ ramp) passed, adding +0.77 dB to FlowDPS-on-RF; the seven failed candidates are documented as diagnosed negative results, including a DPS-vs- Π -GDM L2-norm/squared convention mismatch and a guidance-frequency-vs-integration-accuracy trade-off in Heun. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators: prefer DPS when the operator is calibrated; consider flow-based posterior sampling when the operator may be mis-specified.

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